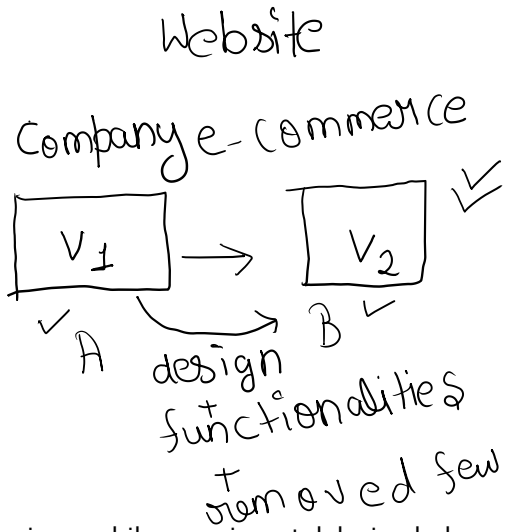
# A/B Testing, Experimental Design & Seasonality and Time Series




A/B testing compares two versions, while experimental design helps make that comparison fair.

## A/B Testing

A/B Testing is an experiment that compares two versions using predifined outcome logic.
Version A: Control or existing version
Version B: Treatment or new version.
Metric: The outcome used to compare the versions

| Industry | Version A | Version B | Metric |
|---|---|---|---|
| E-commerce | Exisiting checkout | New checkout | Purchase conversion Rate |
| Marketing | Email Subject A | Email Subject B | Click Through Rate. |
| Banking | Exiting App Form | Simplified Form | Completion Rate |
| Education | Existing course Page | Redesigned Page | Enrollment Rate. |

**Calcualte conversion Rates**

| Metric | Version A | Version B |
|---|---|---|
| Visitors | 1,000 | 1,000 |
| Purchases | 100 | 130 |

Formula :

$$CR = \frac{\text{Conversions}}{\text{Visitors}} \times 100$$

Version A:

=100/1000*100 10.00%

Version B:

=130/1000*100 13%

Version B has higher observed conversion rate. We now need to assess whther the difference is statistically significant.

**Absolute and Relative Lift**

Absolute Lift:

pB - pA =13-10 = 3 percentage points

Relative Lift:

(pB-pA)/pA *100

=13-10/10*100

30%

**Interpretation:**

Absolute lift = 3 percentage points

Relative Lift = 30%

**Formulate the Hypothesis:**

For two tailed test:

**Null Hypothesis:**

Ho: pA = pB

There is no difference between the popualtion conversion rates.

**Alternative Hyptothesis:**

H1: pA ≠ pB

The population conversion rates differs.

Choose the significance level

$$\alpha = 0.05$$

Calculating critical Z value for two tailed test with alpha = 0.05

$$Z_{critical} = \pm 1.96$$

**Decision Rule**

If Z < -1.96, reject Ho

If Z > 1.96, Reject Ho

If -1.96 ≤ Z ≥ +1.96, Failed to reject Ho

**Two Proportion Z-Test using Critical values**

For two indpendent conversion proprtions, under the null hypothesis that the proportions are equal, claculate the pooled proportion:

$$\hat{p} = \frac{x_A + x_B}{n_A + n_B}$$

SE under the null:

$$SE_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}$$

**Test Statistic:**

$$Z = \frac{\hat{P}_B - \hat{P}_A}{SE_0}$$

Where:
xA,xB: Number of conversions
nA,nB: number of visitors
$\hat{p}A, \hat{p}B$: observed conversion proportions.

**Example:**

| Metric | Version A | Version B |
|---|---|---|
| Visitors | 1,000 | 1,000 |
| Purchases | 100 | 130 |
| CR | 0.1 | 0.13 |

**Step 1: Calculated the pooled proprtion:**

$$\hat{P} = \frac{100+130}{1000+1000} = \frac{230}{2000}$$

0.115

**Step 2: Calculate Standard Error:**

$$SE_0 = \sqrt{0.115(1-0.115)\left(\frac{1}{1000}+\frac{1}{1000}\right)} \qquad \frac{1}{1000} = 0.001$$

$$= \sqrt{0.115(0.885)(0.002)}$$

$$= \sqrt{0.000204}$$

$$SE_0 = 0.014267$$

**Step 3: Calculate the Z statistic:**

Z = 0.13-0.10/0.014267

$Z = 2.102741$

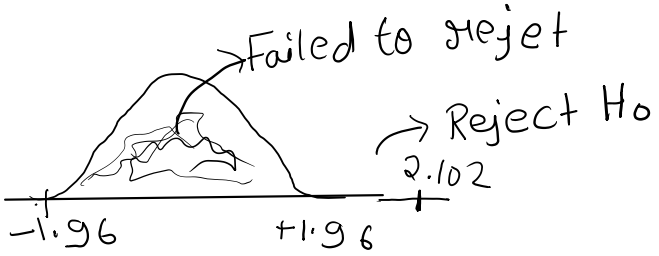
**Step 4: Compare the critical value**

For two tailed test at the 5% significance level:

Zcritical = +-1.96

Z = 2.1027

2.102>1.96




Decision : Reject Null Hypothesis.

**Step 5: Write the businesss Conclusion**

The observed conversion rate differs significantly at 5% level under the test assumptions. But Version B has the higher observed conversion rate : 13% compared to 10%

This provides statistical evidence of a difference. It does not guarantee that the improvement will persist or prove commercial profitability.

**One tailed vs Two tailed tests**

| Test | Alternative Hypothesis | Critcal value at $\propto$ =0.05 |
|---|---|---|
| 2 tailed | pA ≠ pB | ±1.96 |
| Right Taile | pB > pA | 1.645 |
| Left tailed | pB < pA | -1.645 |

Choose the direction before analysing the results. Don't switch to a one tailed test after seeing the result.

## Experimental Design Prenciples

Experimental design is the process of planning how participants, treatements, measurements and comparisons will be arranged to answer a research question reliably.

**Four important principles**

1. Radomization: Assign eligible participants randomly A or B.
2. Control Group: Keep an existing version as a baseline comparison
3. Consistent Measurement: Use the same metric definition and observation window for both the groups.
4. Adequate sample Size: Plan enough observation to detect a meaningful effect with suitable statistical power.

**Practical Business case:**

An online food delivery company wants to reduce checkout abandonment.

| **Component** | **Experimental design** |
|---|---|
| Control group | Existing checkout |
| Treatment group | Simplified checkout |
| Randomization unit | Eligible user |
| Primary metric | Order completion rate |
| Guardrail metric | Payment failure rate |
| Observation window | Predefined period |
| Analysis | Compare completion proportions |
| Decision | Evaluate statistical evidence and business impact |

**Why Randomize?**
If returning customer receive Version B and new customer receives version A, difference may reflect customer experience rather than checkout design.
It helps reduce systematic difference between th groups.

**Common Experimental Mistake:**
1. Changing ht primary metric after seeing the results.
2. Stopping early when the result looks favorable.
3. Assigning different customer types to each version.
4. Ignoring sample size and experiment duration.
5. Testing multiple variations without considering multiple comparisons.
6. Ignoring the guardrail metrics or unexpected events.

**Example:**
Assume:
Monthly eligible vistors = 20,000
Baseline conversion rate = 10%
New CR = 13%
Contribution Margin per additional conversion = ₹200.
Monthly redisgn cost = ₹50,000

Estimated additional conversions = 20,000 *(0.13-0.10) = 20,000*0.03 = 600
Estimated additional contribution = 600* ₹200 = ₹1,20,000
Estimated net benefit= ₹1,20,00-₹50,000 = ₹70,000

**Interpretation:**
The estimated monthly net benefit is ₹70,000 if the observed conversion lift persists and the assumptions hold.
The business should also consider uncertainty, implementation risks and gurdrail metrics before rolling out the change.

## Seasonaloty & Time Series

**What is Time series data?**
Time-series data consists of observartions recorded in chronological order, usually at regular intervals.
Example:
Daily orders
Weekly website traffic
Monthly Revenue
Quarterly Profits
Annual Customer churn

**Four components**
1. Trend: Long term direction
Example: Revenue grows over several years
2. Seasonality: Repeating pattern at known frequency.
Example: Retail sales rise every December/particular festival

3. Cycilical Movement: Longer term fluctuation without fixed period.
Example: Demand changes with economic conditions .
4. Irregular Variation: Unpredicted fulctuation
Example: Unexpected supply disrutption.

**Practical Dataset**

| Quarter | Sales |
|---|---|
| Q1, Year 1 | 100 |
| Q2, Year 1 | 120 |
| Q3, Year 1 | 110 |
| Q4, Year 1 | 150 |
| Q1, Year 2 | 110 |
| Q2, Year 2 | 132 |
| Q3, Year 2 | 121 |
| Q4, Year 2 | 165 |

1. Did the total sales increase?
2. Which quarter has the highest sales?
3. Is a seasonal pattern possible?
4. Is two years quarterly data enough to establish seasonality confidently?

**Year 1 total:**
480
**Year 2 Total:**
528

**Year Over Year Growth:**

$$\frac{528-480}{480} \times 100 = 10.00\%$$

Q4 is the highest quarter in both the years, suggesting a possible seasonal pattern. However two years data are insufficient to establish a reliable recurring pattern.

$$\%\ Y\ growth = \frac{Curr\ PD - Prie\ Yr\ PD}{Prie\ Yr\ PD} \times 100$$

**Interpreatation:**
The dataset shows consistent year-over year growth while maintiining the same relative quarterly pattern.

**<u>Additive vs Multiplicative Models</u>**

**Additive Model:**

$$Y = T + S + I$$

Useful when seasonal changes are roughly constant in asolute size.
Example:
Sales increase by about ₹20 lakhs every December.

**Multiplicative Model:**

$$Y = T * S * I$$

Useful when seasonal changes are proportional to the underying sales level.

**Example:**

December sales are consistent about 30% higher than ordinary months

**Basic Forecasting**

**Naïve Forecast**

$$Y_{t+1} = Y_t$$

The nextr forecast equals the latest observed value

**Seasonal Naïve Forecast**

$$Y_t = Y_{t-m}$$

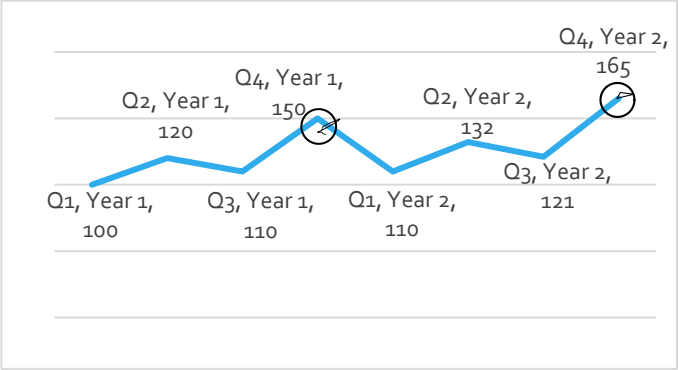
Here M is the numbder of periods in seasonal cycle.

Example:

For quarterly data use m = 4 to repeat the corresponding quarter from the previous year.

These are baseline methods that can be compared against more advanced forecasts.

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|---|---|
| Q1, Year 1 | 100 |
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**Why is randomization important?**

It helps reduce systematic differences between experimental groups.

**What is the difference between trend and seasonality?**

Trend describes long-term direction; seasonality describes recurring patterns.

**What is Seasonal Naïve forecast?**

A forecast based on the value from the corresponding period in the previous seasonal cycle.

**What is the two tailed critical Z value at 5% significance level?**

Approximately ±1.96.